MSG No. 51.301 P_Amma [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `.2/m.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,7]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,4,6,8]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,11,13,15]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,12,14,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `.2/m.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,3,5,7]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,4,6,8]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[9,11,13,15]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[10,12,14,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[2,3,5,8]`
3	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,13,16]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,13]`
2	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,14]`
3	$\left[0,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,15]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,16]`
5	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,9]`
6	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[6,10]`
7	$\left[0,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[7,11]`
8	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[8,12]`

Table 5: Wyckoff site: `8e`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3]`
2	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[2,4]`
3	$[0,\ -y,\ 0]$	`[5,7]`
4	$\left[\frac{1}{2},\ y,\ 0\right]$	`[6,8]`

continued ...

Table 5

No.	position	mapping
5	$\left[0,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[9,11]
6	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[10,12]
7	$\left[0,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[13,15]
8	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[14,16]

Table 6: Wyckoff site: 8f, site symmetry: .m.

No.	position	mapping
1	$[x,\, 0,\, z]$	[1,7]
2	$\left[x+\frac{1}{2},\, 0,\, -z\right]$	[2,8]
3	$[-x,\, 0,\, -z]$	[3,5]
4	$\left[\frac{1}{2}-x,\, 0,\, z\right]$	[4,6]
5	$\left[x,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[10,16]
7	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[11,13]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[12,14]

Table 7: Wyckoff site: 8g, site symmetry: m..

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, z\right]$	[1,6]
2	$\left[\frac{3}{4},\, -y,\, -z\right]$	[2,5]
3	$\left[\frac{3}{4},\, y,\, -z\right]$	[3,8]
4	$\left[\frac{1}{4},\, -y,\, z\right]$	[4,7]
5	$\left[\frac{1}{4},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9,14]
6	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10,13]
7	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11,16]
8	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[12,15]

Table 8: Wyckoff site: 16h, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[2]
3	$[-x,\, y,\, -z]$	[3]
4	$\left[\frac{1}{2}-x,\, -y,\, z\right]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[6]
7	$[x,\, -y,\, z]$	[7]
8	$\left[x+\frac{1}{2},\, y,\, -z\right]$	[8]
9	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]

continued ...

Table 8

No.	position	mapping
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[14]
15	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16]